

School of Technology Management and Engineering
NMIMS, Chandigarh Campus



WEB PROGRAMMING (WP)

PROJECT FILE

Project Title: Smart Event Management System

Submitted To:

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Submitted By:

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I am extremely thankful to her for providing me with the opportunity to explore, learn, and apply the concepts in a practical way. Her continuous support has motivated me to complete this work with sincerity and dedication.

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Lastly, I extend my sincere appreciation to my classmates, friends, and family members for their encouragement, valuable suggestions, and cooperation throughout the preparation of this work. Without their support, this project would not have been possible.

ABSTRACT

The Smart Event Management System is a web-based application designed to manage and organize college events efficiently. In many institutions, event registration, coordination and certificate distribution are handled manually, which often leads to data mismanagement, duplication and lack of transparency. This system aims to digitize and automate the entire event management process.

The system includes three major modules: Student, Coordinator and Admin. Students can register, view available events, participate in solo or team events and download certificates. Coordinators can create and manage events, track registrations and update event status. The Admin oversees users, categories and overall system management.

For database design, an ER Diagram has been created including entities such as User, Event, Categories, Registration, Team member and Certificates. Proper relationships such as one-to-one and one-to-many have been implemented to maintain data integrity.

Phase 1 of the project focuses on frontend page development and database design, including ER Diagram, relational schema, table creation and insertion of sample data.

PROBLEM STATEMENT

Colleges and universities conduct numerous academic, cultural, technical, and sports events throughout the academic year. These events involve multiple stakeholders including students, coordinators, volunteers, and administrators. However, in many institutions, event management activities such as registration handling, team formation, participation tracking, attendance recording, and certificate distribution are managed manually or through basic tools like spreadsheets and online forms.

Such unstructured systems lead to several challenges including data redundancy, lack of centralized storage, difficulty in tracking student participation history, inefficient coordination between departments, and delays in generating accurate reports. Moreover, retrieving event-related information or analyzing participation trends becomes time-consuming and prone to errors.

There is a need for a structured, centralized, and database-driven system that can efficiently manage the complete event lifecycle. The system should ensure proper data organization, maintain relational integrity, reduce manual workload, and provide reliable access to event and participant information. Therefore, this project aims to design and develop a Smart Event Management System that digitally manages event creation, registration processes, and participation records using proper database management principles.

INTRODUCTION

1. Background

In educational institutions, events play a significant role in enhancing student skills, encouraging participation, and promoting collaboration. These events may include technical competitions, cultural festivals, workshops, seminars, and sports activities. Managing such events requires proper coordination between students, coordinators, and administrators.

However, in many colleges, event-related tasks such as registration handling, team member tracking, participation monitoring, and certificate issuance are performed manually or using unstructured tools. This often results in data inconsistencies, duplication of records, poor tracking of student involvement, and difficulty in generating reports.

2. Motivation

The motivation behind this project is to design a structured and efficient system that eliminates manual errors and provides centralized control over event management activities. By implementing proper database concepts such as entities, relationships, primary keys, and foreign keys, the system ensures data integrity and reliability.

Additionally, a web-based interface makes the system accessible and user-friendly for students, coordinators, and administrators.

3. Objectives

The main objectives of the Smart Event Management System are:

- To design a normalized relational database for event management
- To develop an ER diagram representing real-world relationships
- To implement tables with primary and foreign key constraints
- To provide a frontend interface for student interaction

- To ensure efficient tracking of registrations and event participation
- To provide automated certificate generation and downloading for participants.
- To allow coordinators to track team registrations and manage team members efficiently.
- To generate various reports such as participant lists, event attendance, and performance summaries.
- To provide a dashboard view for admins to monitor overall event activities.
- To ensure secure user authentication and data privacy for all participants.

FRONTEND

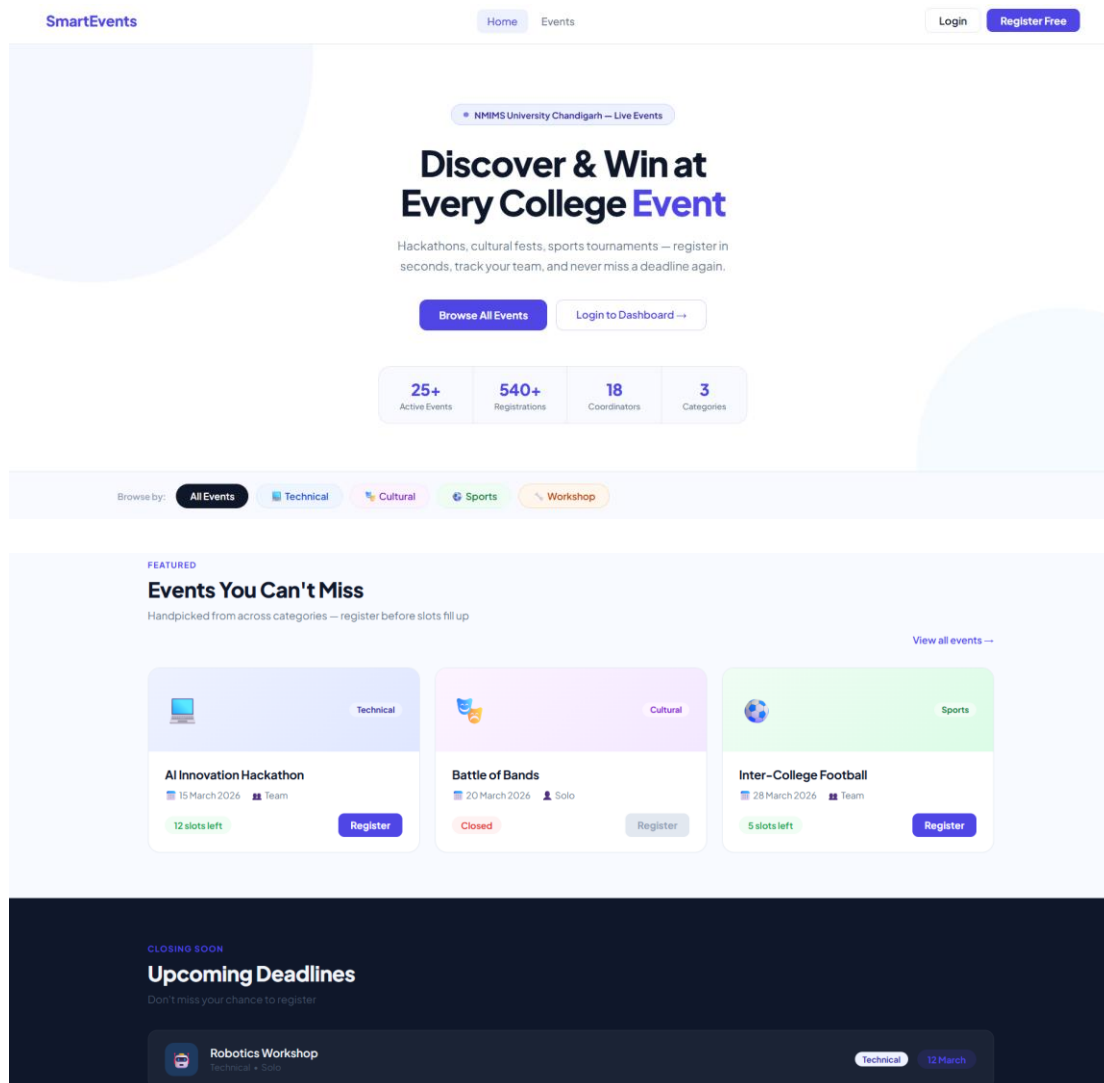
Progress details / Modules completed

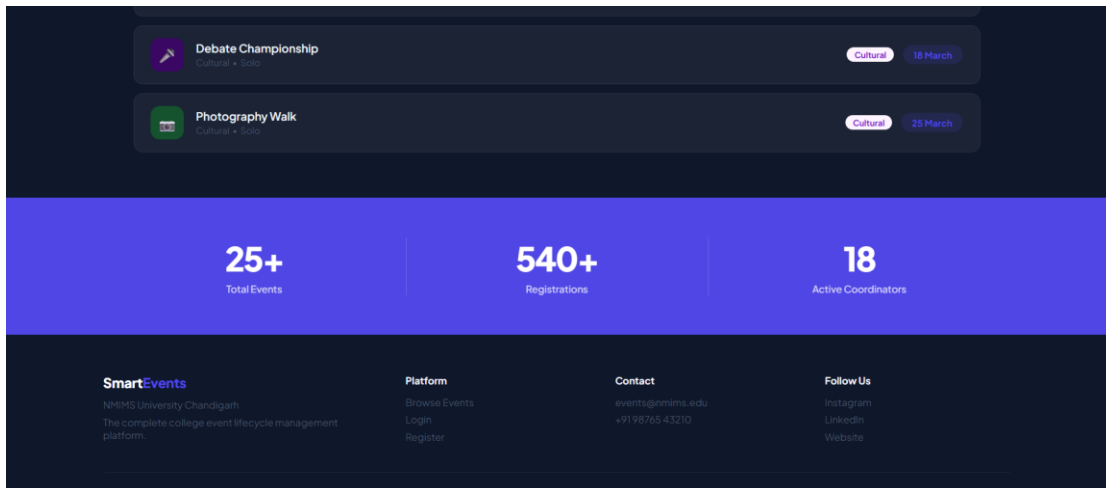
The Smart Event Management System has been divided into three main modules: Student Module, Coordinator Module, and Admin Module.

1. Student Module

a) Public pages (before login):

- Homepage – Project introduction, featured events, upcoming events, quick stats and navigation.





- Login / Register Page – User authentication with email, phone, and Google sign-in.

SmartEvents

— Back to Home

NMIMS University Chandigarh

Welcome back to SmartEvents

Your one-stop platform for discovering, registering and tracking all college events.

- Discover Events**
Browse hackathons, tests and sports
- Register Instantly**
One-click registration with your team
- Track Participation**
View your history and certificates

25+ Events 540+ Students 18 Coordinators

Student Coordinator Admin

Login Register

Email Address
you@nmims.edu

Password
Enter your password [Forgot password?](#)

Login to Dashboard

or

Continue with Google

NMIMS University Chandigarh

Welcome back to SmartEvents

Your one-stop platform for discovering, registering and tracking all college events.

- Discover Events**
Browse hackathons, tests and sports
- Register Instantly**
One-click registration with your team
- Track Participation**
View your history and certificates

25+ Events 540+ Students 18 Coordinators

Student Coordinator Admin

Login Register

Full Name
Your full name

Email Address
you@nmims.edu

Roll Number
e.g. 0028

Password
Create a password (min 6 chars)

Confirm Password
Confirm your password

Phone Number
Enter your phone number

Department

Data Science ☐

Computer Science ☐

Mechanical ☐

Electronics ☐

BBA ☐

☐ I agree to the [Terms & Conditions](#)

- Event Listing Page – Shows all available events with the option to register and filters option.

SmartEvents Home **Events** Login Register Free

EXPLORE

All Events

Discover technical, cultural and sports events at NMIMS Chandigarh

Search events... All Categories All Types All Status Sort: Latest

Showing 3 events

HackNova 2026

12 March 2026 Team (2-4) ₹500/team

24-hour hackathon to build innovative solutions for real-world problems.

12 slots left View Details

Rhythm Fest 2026

20 March 2026 Solo ₹200/person

Dance, music and drama competition on the grand open-air stage.

Registration Closed View Details

Inter-College Football

5 April 2026 Team (11) ₹1000/team

Battle it out on the field – inter-college football championship.

5 slots left View Details

Robotics Workshop

12 March 2026 Solo Free

Hands-on robotics session for beginners and intermediate level students.

Completed View Details

SmartEvents NMIMS University Chandigarh

Platform
Browse Events
Login

Contact
events@nmims.edu
+9198765 43210

Follow Us
Instagram
LinkedIn

- Events description page – Event description, registration fee, rules and requirements, date, time, venue, slots available, coordinator details, etc.

HackNova 2026

24-Hour National Level Hackathon

📅 12 March 2026

👥 Team Event

💰 ₹500 / team

🟢 12 Slots Left

About the Event

HackNova 2026 is a 24-hour national-level hackathon where developers, designers, and innovators collaborate to build impactful technology solutions for real-world problems. This is your chance to innovate, collaborate, and compete with the best minds.

Rules

- Teams must have 2-4 members.
- Only original ideas are allowed — no pre-built projects.
- Plagiarism results in immediate disqualification.
- All development must happen during the event window.
- Judge's decision is final.

Requirements

- Laptop and valid college ID required for all members.
- Teams must report 30 minutes before start time.
- High-speed internet access will be provided.

[Register Now →](#)

Event Details

Date
12 March 2026

Time
10:00 AM – 10:00 AM

Venue
Innovation Lab

Fee
₹500 / team

Type
Team Event

Slots Left
12

AS

Ananya Sharma
Event Coordinator
📞 +91 98765 43210

Rhythm Fest 2026

Dance • Music • Drama • Expression

📅 20 March 2026

👤 Solo / Group

💰 ₹200 / participant

🔴 Registration Closed

About Rhythm Fest

Rhythm Fest is the cultural heartbeat of NMIMS, celebrating creativity through dance, music, and drama. Participants showcase talent, emotion, and storytelling on a grand open-air stage. This is the biggest cultural event of the semester.

Rules & Requirements

- Solo and group performances both allowed.
- Time limit: 4-6 minutes per performance.
- Props allowed — to be arranged by participants.
- Obscenity or offensive content leads to disqualification.
- Participants must report 1 hour before event time.
- College ID mandatory.

Registration Closed

Event Details

Date
20 March 2026

Time
5:00 PM – 9:00 PM

Venue
Open Air Auditorium

Fee
₹200 / person

Type
Solo / Group

Status
Closed

RM

Riya Mehta
Event Coordinator
📞 +91 91234 56789

Inter-College Football Championship

Where passion meets performance

5 April 2026

Team (11+3)

₹1000 / team

5 Slots Left

About the Event

The Inter-College Football Championship brings together top teams from various colleges to compete for glory, sportsmanship, and pride. This tournament promotes teamwork, discipline, and the competitive spirit of college sports.

Rules

- Each team must have 11 players + 3 substitutes.
- Standard FIFA rules will be followed throughout.
- Red card results in immediate suspension from the tournament.
- Referee's decision is final and cannot be challenged.
- Teams must report 30 minutes before match time.

Requirements

- College ID mandatory for all players.
- Proper sports kit and studs required.
- Team captain must carry the registration confirmation.

[Register Now →](#)

Match Details

Date
5 April 2026

Time
9:00 AM onwards

Venue
College Main Ground

Fee
₹1000 / team

Type
Team Event

Slots Left
5 Teams

RM **Rahul Mehta**
Sports Coordinator
+91 91234 56789

b) Student Dashboard (after login):

- Student Dashboard – Overview of registered events, upcoming events, participations, and upcoming deadlines.

SmartEvents
Dashboard
Events
My Events
Profile
Logout

Welcome back, Ashna 🎉
Here's a quick overview of your activity
Browse Events →

3
Registered Events

2
Upcoming Events

5
Total Participations

2
Certificates Earned

Upcoming Deadlines

View all →

- HackNova 2026 Technical 12 Mar
- Rhythm Fest Cultural Closed
- Inter-College Football Sports 5 Apr

My Recent Events

View all →

- HackNova 2026 Team Event • Technical Upcoming
- Rhythm Fest Solo • Cultural Attended
- Football Championship Team Event • Sports Upcoming

Quick Actions

[Browse Events](#)
[My Registrations](#)
[Edit Profile](#)
[Register for Event](#)

- Student Profile Page – Student personal details, stats, recent activity, and option to edit profile and participation history.

SmartEvents Dashboard Events My Events **Profile** Logout

My Profile

Your personal details and participation history Edit Profile

Aditya
B.Tech Data Science • Sem 4
NMIMS University Chandigarh

6 Events 3 Certificates 2 Upcoming

Personal Information

FULL NAME Aditya	EMAIL ADDRESS aditya@nmims.in
PHONE NUMBER 3454657932	ROLL NUMBER N/A
DEPARTMENT Data Science	UNIVERSITY NMIMS Chandigarh

Recent Activity

	HackNova 2026 Registered for team hackathon	Upcoming
	RhythmFest 2026 Participated in cultural fest	Certificate Earned
	Sports Meet 2025 Participated in sports event	Winner 🏆

- Event registration Page – Registration form and team details.

[← Back to Event](#) 12 Slots Left

Register for HackNova 2026

Team Size: 2-4 members • Fee: ₹500 / team

Team Registration

Fill in your team details to complete registration

Team Name *

Team Members

Member 1 — Team Leader *

Member 2 *

Member 3 (Optional)

Member 4 (Optional)

☐ I have read and agree to the event rules and code of conduct

Confirm Registration → Cancel

Event Summary

Event
HackNova 2026

Date
12 March 2026

Time
10:00 AM – 10:00 AM (24hr)

Venue
Innovation Lab

Fee
₹500 / team

Slots Left
12

AS
Ananya Sharma
Event Coordinator
📞 +9198765 43210

- My Events page – Registered events, attendance status and registration status

SmartEvents Dashboard Events **My Events** Profile Logout

My Registered Events

Track your participation, attendance and certificates

Search events ... [+ Register for Event](#)

Event Name	Type	Category	Registration	Attendance	Certificate	Action
HackNova 2026 12 March 2026	Team	Technical	Registered	Upcoming	Pending	View →
Rhythm Fest 2026 20 March 2026	Solo	Cultural	Closed	Attended	Earned ✓	View →
Inter-College Football 5 April 2026	Team	Sports	Registered	Upcoming	Pending	View →

2. Coordinator/Volunteer Module

- Dashboard

SmartEvents Dashboard **My Events** Participants Attendance Profile Logout Coordinator

Welcome back, Ananya 🌟
Here's an overview of your assigned events and pending actions

[View My Events →](#)

4
Assigned Events

87
Total Participants

3
Pending Actions

2
Completed Events

Assigned Events [View all →](#)

- HackNova 2026
Technical • 12 March 2026 [Upcoming](#)
- Rhythm Fest 2026
Cultural • 20 March 2026 [Closed](#)
- Football Championship
Sports • 5 April 2026 [Upcoming](#)
- Robotics Workshop
Workshop • 12 March 2026 [Completed](#)

Pending Actions [3 pending](#)

- Mark Attendance
Robotics Workshop – overdue [Do Now](#)
- Review Participants
HackNova 2026 – 12 new [Review](#)
- Update Instructions
Rhythm Fest – venue changed [Update](#)

Slot Utilization Across your assigned events

HackNova 2026	38 / 50	Rhythm Fest 2026	30 / 30
Football Championship	9 / 14	Robotics Workshop	20 / 20

- My events page

SmartEvents Dashboard **My Events** Participants Attendance Profile Logout Coordinator

Assigned Events

All events you are responsible for managing

All Status ▾

Technical

HackNova 2026
12 March 2026 • Innovation Lab • 38 / 50 registered

[Manage Event](#) [View Participants](#) [Attendance](#)

Cultural

Rhythm Fest 2026
20 March 2026 • Open Air Auditorium • 30 / 30 registered

[Manage Event](#) [View Participants](#) [Attendance](#)

Sports

Football Championship
5 April 2026 • College Main Ground • 9 / 14 teams registered

[Manage Event](#) [View Participants](#) [Attendance](#)

Workshop

Robotics Workshop
12 March 2026 • Lab 3, Block B • 20 / 20 attended

[View Details](#) [Participants](#) [View Attendance](#)

○ Participants page

SmartEvents

DashboardMy EventsParticipantsAttendanceProfileLogout

Coordinator

Participant Management

View and manage participants across your assigned events

HackNova 2026

38
Total Registered

12
Slots Remaining

10
Teams Registered

Search by name or roll number...

All Types

All Status

Participants – HackNova 2026

Showing 5 participants

#	Participant / Team	Roll No.	Type	Members	Registered On	Status	Action
1	Team CodeCrush Ashna Bansal (Leader)	22BCE1234	Team	4	8 Mar 2026	Confirmed	View
2	Team ByteForce Rohan Sharma (Leader)	22BCE5678	Team	3	9 Mar 2026	Confirmed	View
3	Team NullPointer Priya Mehta (Leader)	22BCE9012	Team	4	10 Mar 2026	Pending	View
4	Team DevHawks Arjun Kapoor (Leader)	22BCE3456	Team	2	10 Mar 2026	Confirmed	View
5	Team StackOverflow Simran Kaur (Leader)	22BCE7890	Team	3	11 Mar 2026	Confirmed	View

Export Participant List

○ Attendance page

Attendance Management

Mark and view attendance for your assigned events

HackNova 2026

Save Attendance

38
Total Registered

0
Marked Present

38
Absent / Unmarked

Attendance Progress

0%

Quick Mark:

Mark All Present

Mark All Absent

#	Participant / Team	Roll No.	Type	Present	Remarks
1	Team CodeCrush Ashna Bansal (Leader)	22BCE1234	Team	☐	Optional remarks...
2	Team ByteForce Rohan Sharma (Leader)	22BCE5678	Team	☐	Optional remarks...
3	Team NullPointer Priya Mehta (Leader)	22BCE9012	Team	☐	Optional remarks...
4	Team DevHawks Arjun Kapoor (Leader)	22BCE3456	Team	☐	Optional remarks...
5	Team StackOverflow Simran Kaur (Leader)	22BCE7890	Team	☐	Optional remarks...

Export Attendance

Save Attendance

○ Profile

My Profile

Your coordinator details and event history

[Edit Profile](#)



Ananya Sharma
Event Coordinator
Computer Science Dept.
[Coordinator](#)

4

Events

87

Participants

2

Completed

Personal Information

FULL NAME Ananya Sharma	EMAIL ananya@college.edu
PHONE +91 98765 43210	DEPARTMENT Computer Science
ROLE Event Coordinator	UNIVERSITY NMIMS Chandigarh

Assigned Roles & Event History

	HackNova 2026 Lead Coordinator • Technical	Upcoming
	Rhythm Fest 2026 Co-Coordinator • Cultural	Closed
	Robotics Workshop Lead Coordinator • Workshop	Completed
	Football Championship Support Coordinator • Sports	Upcoming

3. Admin Module

○ Dashboard

SmartEvents Admin

Dashboard Events Users Roles Analytics Attendance Settings Logout

Admin

Admin Dashboard

Full system overview — NMIMS University Chandigarh

[+ Create Event](#) [View Analytics](#)



25
Total Events
+ 4 this month



540
Total Users
+ 38 this week



1,240
Registrations
+ 12% vs last month



18
Coordinators
+ 2 newly assigned

Registrations This Month

This Month



Total this period: 437 registrations

Events by Category

All time



25

- Technical — 10 (40%)
- Cultural — 8 (32%)
- Sports — 7 (28%)

Recent Events

Manage all →

	HackNova 2026 Technical • 38/50 registered	Upcoming Edit
---	--	---------------

Coordinator Workload

Manage →

AS	Ananya Sharma	3 events
----	---------------	----------

○ Events page

Event Control

Create, update and delete events across all categories

[+ Create New Event](#)[Export](#)

All Categories

All Status

Sort: Latest

[Clear](#)

Showing 6 events

#	EVENT NAME	CATEGORY	DATE	TYPE	FEE	SLOTS	REGISTERED	STATUS	ACTIONS
1	Hacknova 2026 Innovation Lab, Block A	Technical	12 Mar 2026	team	₹500.00	20	0	Upcoming	Edit Delete
2	Rhythm fest Auditorium hall	Cultural	20 Mar 2026	solo	₹200.00	50	0	Upcoming	Edit Delete
3	Inter-college football Sports ground	Sports	5 Apr 2026	team	₹300.00	16	0	Upcoming	Edit Delete
4	Campus stars Auditorium	Cultural	13 Apr 2026	Team	₹100.00	30	0	Upcoming	Edit Delete
5	Chess tournament Common room	Sports	25 Apr 2026	Both	Free	30	0	Upcoming	Edit Delete
6	Nachbaliyeee Near flap barriers	Cultural	13 May 2026	Both	₹200.00	30	0	Upcoming	Edit Delete

Users page

User Management

Manage students, coordinators and their access

[Students \(522\)](#)[Coordinators \(18\)](#)**522**

Total Students

**18**

Coordinators

**540**

Total Active Users

All Departments

All Status

[Clear](#)

Students

Showing 5

#	STUDENT	ROLL NO.	DEPARTMENT	EVENTS	STATUS	JOINED	ACTIONS
1	AB Ashna Bansal ashna@nmims.edu	22BCE1234	Data Science	3	Active	Jan 2026	View Disable
2	RS Rohan Sharma rohan@nmims.edu	22BCE5678	Computer Science	5	Active	Jan 2026	View Disable
3	PM Priya Mehta priya@nmims.edu	22BCE9012	MBA	2	Inactive	Feb 2026	View Enable
4	AK Arjun Kapoor arjun@nmims.edu	22BCE3456	Computer Science	4	Active	Jan 2026	View Disable
5	SK Simran Kaur simran@nmims.edu	22BCE7890	Data Science	3	Active	Jan 2026	View Disable

Roles page

Role Assignment

Assign coordinators to events and manage their roles

[+ Assign Coordinator](#)

Events Needing Coordinator

3 unassigned



HackNova 2026
Technical • 12 Mar 2026

[Assign →](#)

Football Championship
Sports • 5 Apr 2026

[Assign →](#)

Debate Championship
Cultural • 18 Mar 2026

[Assign →](#)

Coordinator Availability

AS

Ananya Sharma
3 events assigned

Busy

RM

Rahul Mehta
2 events assigned

Busy

RK

Riya Kapoor
4 events assigned

Overloaded

VT

Vikram Taneja
1 event assigned

Available

All Categories ▾

Current Assignments

Total: 2

EVENT	CATEGORY	DATE	COORDINATOR	ROLE TYPE	STATUS	ACTION
Rhythm Fest 2026	Cultural	20 Mar 2026	RK Riya Kapoor	Lead Coordinator	Closed	Remove
Robotics Workshop	Workshop	12 Mar 2026	RM Rahul Mehta	Support Coordinator	Completed	Remove

Analytics page

Participation Analytics

Data-driven insights on student involvement and event effectiveness

This Month ▾

[Export Report](#)

437

Total Registrations
↑ 18% vs last month



84%

Avg Attendance Rate
↑ 6% improvement



HackNova

Most Popular Event

76% slot utilization

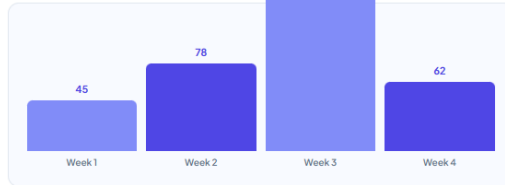


2.3

Avg Events/Student
↑ from 1.8 last sem

Monthly Registrations Trend

2026



Participation by Category

- Technical — 219 (50%)
- Cultural — 131 (30%)
- Sports — 87 (20%)

All Categories ▾

Sort: Utilization ▾

Event Performance Report							
EVENT	CATEGORY	SLOTS	REGISTERED	ATTENDED	SLOT UTILIZATION	ATTENDANCE RATE	STATUS
Rhythm Fest 2026	Cultural	30	30	28	100%	93%	Closed
Robotics Workshop	Workshop	20	20	20	100%	100%	Completed
Debate Championship	Cultural	40	35	32	88%	91%	Completed
HackNova 2026	Technical	50	38	—	76%	Upcoming	Upcoming
Football Championship	Sports	14	9	—	64%	Upcoming	Upcoming

Most Active Students					By number of events participated
 Rohan Sharma 5 events 	 Arjun Kapoor 4 events 	 Ashna Bansal 3 events 	 Simran Kaur 3 events 	 Priya Mehta 2 events 	

Attendance page

Attendance Overview

All Events

Export All

Event-wise attendance data across all modules

94%

Overall Attendance Rate

↑ 6% vs last semester

80

Total Attendees (Completed)

2

Pending Attendance

Needs coordinator action

Robotics Workshop

12 Mar 2026 • Workshop

Completed

Attendance Rate

100% (20/20)

20

Present

0

Absent

20

Total

View Detailed Attendance →

Rhythm Fest 2026

20 Mar 2026 • Cultural

Completed

Attendance Rate

93% (28/30)

28

Present

2

Absent

30

Total

View Detailed Attendance →

Debate Championship

18 Mar 2026 • Cultural

Completed

Attendance Rate

91% (32/35)

32

Present

3

Absent

35

Total

HackNova 2026

12 Mar 2026 • Technical

Upcoming

Attendance will be available after the event on 12 Mar 2026

38

Registered

12

Slots Left

Settings page

System Settings

Configure categories, limits and platform preferences

[Reset to Defaults](#)[Save All Settings](#)

Event Categories

 **Technical**
10 events • Active

[Disable](#)[Remove](#)

 **Cultural**
8 events • Active

[Disable](#)[Remove](#)

 **Sports**
7 events • Active

[Disable](#)[Remove](#)

 **Workshop**
3 events • Active

[Disable](#)[Remove](#)[+ Add](#)

Registration Limits

Max Events Per Student

Default Max Team Size

Default Min Team Size

Registration Deadline (days before event)

Max Coordinators Per Event

Platform Preferences

Email Notifications

Send email confirmations on registration and updates



Auto Waitlist

Automatically add students to waitlist when slots are full



Public Event Listing

Allow events to be visible without login



Certificate Generation

Auto-generate certificates for attended events



Dark Mode (System-wide)

Enable dark mode across all dashboard pages



Maintenance Mode

 Students cannot register during maintenance



Event Status Control

Override status for any event

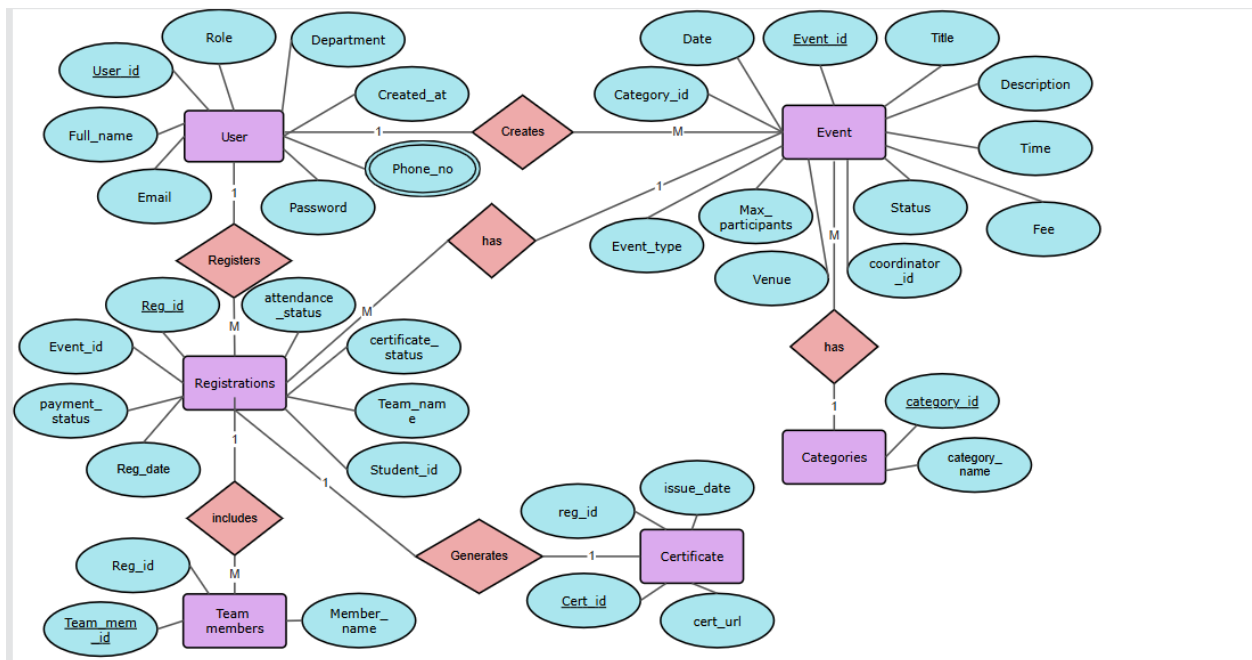
EVENT	CURRENT STATUS	CHANGE TO	APPLY
HackNova 2026	Upcoming	Upcoming ▼	Apply
Rhythm Fest 2026	Closed	Closed ▼	Apply
Football Championship	Upcoming	Upcoming ▼	Apply
Robotics Workshop	Completed	Completed ▼	Apply

BACKEND

1. Database Design

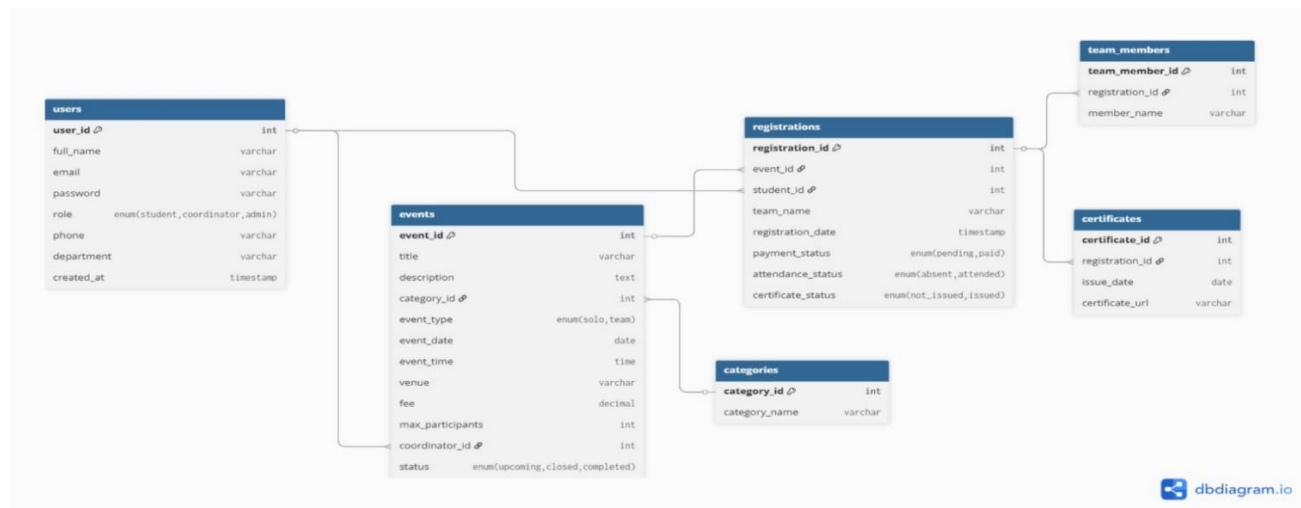
a) ER Diagram:

- Entities identified: User, Event, Category, Registration, Team Member, Certificate
- Relationships defined:
 - One Coordinator (User) creates many Events
 - One Student (User) can register for many Events
 - One Event can have many Registrations
 - One Registration can have many Team Members
 - One Registration can have one Certificate
 - One Category can have many Events



b) Relational Schema:

- Tables created or schema created: users, categories, events, registrations, team_members, certificates
- Primary and foreign keys defined according to ER diagram relationships.
- Constraints such as UNIQUE (for email), ENUMs (for role, event type, and status), and NOT NULL added to maintain data integrity.



c) Tables and Sample Data:

- Inserted sample records for all tables to demonstrate functionality during Phase 1 evaluation.
- Includes sample students, events, categories, registrations, team members, and certificates.
- Constraints such as UNIQUE (for email), ENUMs (for role, event type, and status), and NOT NULL added to maintain data integrity.

```

mysql> select * from users;
+-----+-----+-----+-----+-----+-----+-----+-----+
| user_id | full_name | email | password | role | phone_no | department | created_at |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Admin user | admin@college.edu | admin1234 | admin | 987654321 | Administration | 2026-02-25 12:02:39 |
| 2 | Ananya sharma | ananya@college.edu | coord1234 | coordinator | 98567894 | Computer Science | 2026-02-25 12:03:36 |
| 3 | Rahul verma | Rahul@college.edu | coord1234 | coordinator | 78645353 | Sports | 2026-02-25 12:04:19 |
| 4 | Ashna Bansal | ashna@email.com | student1234 | student | 78890346 | Data science | 2026-02-25 12:05:10 |
| 5 | Riya Kapoor | riya@email.com | student1234 | student | 784543436 | BBA | 2026-02-25 12:05:45 |
| 6 | Arjun Mehta | arjun@email.com | student1234 | student | 89543436 | Mechanical | 2026-02-25 12:17:17 |
| 7 | Karan Patel | karan@email.com | student1234 | student | 999543436 | Electronics | 2026-02-25 12:18:04 |
+-----+-----+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
  
```

```
mysql> select * from categories;
+-----+-----+
| c_id | c_name |
+-----+-----+
| 1 | Technical |
| 2 | Cultural |
| 3 | sports |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from events;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| event_id | title | description | c_id | event_typt | event_date | event_time | venue | fee | max_ |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Hacknova 2026 | 24-hour ai innovation hackathon | 1 | team | 2026-03-12 | 10:00:00 | Main innovation lab | 500.00 | 20 |
| 2 | Rhythm fest | Annual music and dance competition | 2 | solo | 2026-03-20 | 16:00:00 | Auditorium hall | 200.00 | 50 |
| 3 | Inter-college football | Football championship | 3 | team | 2026-04-05 | 09:00:00 | Sports ground | 300.00 | 16 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from registrations;
+-----+-----+-----+-----+-----+-----+-----+-----+
| reg_id | event_id | student_id | team_name | reg_date | payment_status | attendance_status | certificate_status |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 4 | Codestorm | 2026-02-25 12:43:30 | paid | absent | not_issued |
| 2 | 2 | 4 | Null | 2026-02-25 12:43:58 | paid | attended | issued |
| 3 | 3 | 5 | Goalgetters | 2026-02-25 12:44:21 | pending | absent | not_issued |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from team_members;
+-----+-----+-----+
| team_mem_id | reg_id | mem_name |
+-----+-----+-----+
| 1 | 1 | Ashna bansal |
| 2 | 1 | Riya Kapoor |
| 3 | 1 | Arjun mehta |
| 4 | 3 | Riya Kapoor |
| 5 | 3 | Karan patel |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from certificates;
+-----+-----+-----+-----+
| cert_id | reg_id | issue_date | cert_url |
+-----+-----+-----+-----+
| 1 | 2 | 2026-03-22 | certificates/rhythmfest_ashna.pdf |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

d)Server.js and terminal

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\SEM4\Smart event management project\Backend> node server.js
Server started
Connected to MySQL ✓
```

```
Incoming Data: {
  full_name: 'vaibhav',
  email: 'vaibhav@gmail.com',
  password: 'vail23',
  role: 'student',
  phone_no: '3456789876',
  department: 'Mechanical'
}
User Inserted: ResultSetHeader {
  fieldCount: 0,
  affectedRows: 1,
  insertId: 14,
  info: '',
  serverStatus: 2,
  warningStatus: 0,
  changedRows: 0
}
```

e)Tables with real data

```
mysql> select * from categories;
+-----+-----+
| c_id | c_name |
+-----+-----+
| 1    | Technical |
| 2    | Cultural |
| 3    | sports |
+-----+-----+
3 rows in set (0.00 sec)
```



```
mysql> select * from certificates;
+-----+-----+-----+-----+
| cert_id | reg_id | issue_date | cert_url |
+-----+-----+-----+-----+
| 1 | 2 | 2026-03-22 | certificates/rhythmfest_ashna.pdf |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from events;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| event_id | title | description | c_id | event_type | event_date | event_time | venue | fee |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Hacknova 2026 | 24-hour ai innovation hackathon | 1 | team | 2026-03-12 | 09:30:00 | Innovation Lab, Block A | 500.00 |
| 2 | Rhythm fest | Annual music and dance competition | 2 | solo | 2026-03-20 | 16:00:00 | Auditorium hall | 200.00 |
| 3 | Inter-college football | Football championship | 3 | team | 2026-04-05 | 09:00:00 | Sports ground | 300.00 |
| 4 | Campus stars | Talent hunt | 2 | Team | 2026-04-13 | 03:00:00 | Auditorium | 100.00 |
| 6 | Chess tournament | A chess tournament | 3 | Both | 2026-04-25 | 20:00:00 | Common room | 0.00 |
| 7 | Nachbaliyeee | Nach ke dikhaooo | 2 | Both | 2026-05-13 | 02:00:00 | Near flap barriers | 200.00 |
| 8 | Prom Night | Prom night | 2 | Team | 2026-04-30 | 06:00:00 | Auditorium | 0.00 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
mysql> select * from users;
+-----+-----+-----+-----+-----+-----+-----+-----+
| user_id | full_name | email | password | role | phone_no | department | created_at |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Admin user | admin@college.edu | admin1234 | admin | 987654321 | Administration | 2026-02-25 12:02:39 |
| 2 | Ananya sharma | ananya@college.edu | coord1234 | coordinator | 98567894 | Computer Science | 2026-02-25 12:03:36 |
| 3 | Rahul verma | Rahul@college.edu | coord1234 | coordinator | 78645353 | Sports | 2026-02-25 12:04:19 |
| 4 | Ashna Bansal | ashna@email.com | student1234 | student | 78890346 | Data science | 2026-02-25 12:05:10 |
| 5 | Riya Kapoor | riya@email.com | student1234 | student | 784543436 | BBA | 2026-02-25 12:05:45 |
| 6 | Arjun Mehta | arjun@email.com | student1234 | student | 89543436 | Mechanical | 2026-02-25 12:17:17 |
| 7 | Karan Patel | karan@email.com | student1234 | student | 999543436 | Electronics | 2026-02-25 12:18:04 |
| 8 | Khyati Sharma | khyatisharma2602@gmail.com | ashna1 | student | 981514247 | Data Science | 2026-04-09 17:57:29 |
| 9 | ashna bansal | ASHNA.BANSAL052@nmims.in | ashna123 | admin | 788902450 | Data Science | 2026-04-09 20:17:57 |
| 10 | Reetpal | riri@nmims.in | riri12 | student | 6987654323 | Data Science | 2026-04-12 23:44:11 |
| 11 | Kunal Punni | kunal@nmims.in | kunal1 | student | 6283213329 | BBA | 2026-04-13 01:01:27 |
| 12 | Avneet Kaur | avneet@gmail.com | avneet | student | 8968722011 | Data Science | 2026-04-13 09:17:17 |
| 13 | Aditya | aditya@nmims.in | ashna1234 | student | 3454657932 | Data Science | 2026-04-13 10:22:17 |
| 14 | vaibhav | vaibhav@gmail.com | vail23 | student | 3456789876 | Mechanical | 2026-04-14 23:43:30 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
mysql> select * from registrations;
+-----+-----+-----+-----+-----+-----+-----+-----+
| reg_id | event_id | student_id | team_name | reg_date | payment_status | attendance_status | certificate_status |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 4 | Codestorm | 2026-02-25 12:43:30 | paid | absent | not_issued |
| 2 | 2 | 4 | Null | 2026-02-25 12:43:58 | paid | attended | issued |
| 3 | 3 | 5 | Goalgetters | 2026-02-25 12:44:21 | pending | absent | not_issued |
| 4 | 1 | 4 | codersssssss | 2026-04-07 21:49:29 | pending | absent | not_issued |
| 5 | 1 | 4 | Codehustlers | 2026-04-07 22:23:15 | pending | absent | not_issued |
| 6 | 1 | 4 | Codehustlers | 2026-04-07 22:23:30 | pending | absent | not_issued |
| 7 | 1 | 4 | Data Dudes | 2026-04-07 22:57:46 | pending | absent | not_issued |
| 8 | 1 | 4 | Data coders | 2026-04-08 21:22:05 | pending | absent | not_issued |
| 9 | 1 | 4 | cd | 2026-04-08 21:29:03 | pending | absent | not_issued |
| 10 | 1 | 4 | Lalalalalaal | 2026-04-09 10:59:31 | pending | absent | not_issued |
| 11 | 1 | 4 | curly hair baddies | 2026-04-09 11:15:56 | pending | absent | not_issued |
| 12 | 1 | 4 | ahshsha | 2026-04-09 12:57:18 | pending | absent | not_issued |
| 13 | 1 | 4 | asdhasdwa | 2026-04-13 10:24:12 | pending | absent | not_issued |
| 14 | 1 | 4 | fastcoders | 2026-04-14 23:36:59 | pending | absent | not_issued |
+-----+-----+-----+-----+-----+-----+-----+-----+
14 rows in set (0.00 sec)
```

```
mysql> select * from team_members;
```

team_mem_id	reg_id	mem_name
1	1	Ashna bansal
2	1	Riya Kapoor
3	1	Arjun mehta
4	3	Riya Kapoor
5	3	Karan patel
6	4	abcd
7	4	pqrs
8	4	hijk
9	5	Akash
10	5	Raghav
11	6	Akash
12	6	Raghav
13	7	Rishabh
14	7	Mohit
15	8	ahj
16	8	bhjj
17	9	pramod
18	9	poonam
19	10	Reetpal
20	10	avineet
21	10	asna
22	11	Avneet Kaur
23	11	Devanshi Sharma
24	12	ashna
25	12	asads
26	13	fgdf
27	13	fgdg
28	14	abc
29	14	lhm

```
29 rows in set (0.00 sec)
```

CONCLUSION

The Smart Event Management System successfully demonstrates the design and implementation of a structured, database-driven solution for managing college events. The project effectively replaces traditional manual processes with a centralized and automated system, thereby reducing data redundancy, improving accuracy, and enhancing overall efficiency.

Through the development of a well-defined database including entities such as Users, Events, Categories, Registrations, Team Members, and Certificates, the system ensures proper data organization and maintains relational integrity using primary and foreign key constraints. The ER diagram and relational schema provide a clear representation of real-world relationships and support efficient data management.

The integration of frontend and backend technologies further enhances the system's functionality. The frontend provides a user-friendly interface for interaction, while the backend handles data processing, storage, and retrieval efficiently. The implementation of three distinct modules—Student, Coordinator, and Admin—ensures role-based access and smooth workflow management within the system.

Students can easily register for events, participate individually or in teams, and access certificates. Coordinators can manage events, track participation, and update event details, while administrators have full control over users, categories, and overall system operations.

Overall, the system achieves its objective of digitizing event management and demonstrates the practical application of database management concepts along with web development technologies. It lays a strong foundation for future enhancements such as real-time notifications, advanced analytics, and improved security features, making it a scalable and efficient solution for modern institutions.